BAYESIAN COMPUTATIONAL INFERENCE
ILLINOIS TECH

WEEK 1

EXAMPLE: medical testing

EVENT A: HAVING THE DISEASE
EVENT B: TESTING POSITIVE

PRIOR $P(A) = 0{,}01$
LIKELIHOOD: $P(B|A) = 0{,}99$
FALSE POSITIVE RATE $= 0{,}05$

MARGINAL: $P(B) = P(B|A)\,P(A) + P(B|A^c)\cdot P(A^c)$
$= (0{,}99)\cdot(0{,}01) + (0{,}05)\cdot(0{,}99)$
$\cong 0{,}0594$

POSTERIOR: $P(A|B) = \dfrac{P(B|A)\cdot P(A)}{P(B)} = \dfrac{(0{,}99)\cdot(0{,}01)}{0{,}0594}$

$\boxed{P(A|B) \cong 0{,}167}$

SAIMOS DO PRIOR DE 0,01 PARA 0,167 COM O TESTE

1% => 16,7% COM UM TESTE POSITIVO.

$P(A|B)$ -> POSTERIOR PROBABILITY. OUR AIM.
$P(B|A)$ -> LIKELIHOOD
$P(A)$ -> PRIOR PROBABILITY, OUR START BELIEF
$P(B)$ -> MARGINAL PROBABILITY

THE PRIOR INFORMS YOUR ESTIMATE OF THE POSTERIOR.

BAYESIAN INFERENCE

1. START WITH A PRIOR DISTRIBUTION, WHICH REPRESENTS INITIAL BELIEFS

2. COLLECT DATA

3. COMPUTE THE LIKELIHOOD: THE PROBABILITY OF OUR DATA GIVEN OUR PRIOR.

4. WE COMPUTE THE MARGINAL PROBABILITY (THIS ENSURES OUR ESTIMATE IS NORMALIZED TO 1)
5. WE COMPUTE BAYES' RULE TO COMPUTE THE POSTERIOR DISTRIBUTION (THIS IS OUR <u>UPDATED</u> BELIEFS AFTER SEENING THE DATA)

WEEK 1: ILLINOIS TECH

EXAMPLE: A IS A FAIR COIN?

	PRIOR
H_0: COIN IS FAIR	0,5
HA: COIN IS BIASED	0,5

APOS (TTT)

PRIOR

0,5 → 0,33

A CRENÇA QUE A MOEDA É HONESTA DIMINUIU COM 3 T'S SEGUIDOS

DATA: TTT (3 TAILS)

$$P(H_0 \mid TTT) = \frac{P(TTT \mid H_0) \cdot P(H_0)}{P(TTT)} = \frac{(1/2)^3 (1/2)}{3/16} = \frac{1}{3}$$

MARGINAL

$$P(TTT) = P(TTT \mid H_0) \cdot P(H_0) + P(TTT \mid HA) \cdot P(HA)$$

$P(TTT \mid H_0)$ → SOB H_0, A MOEDA É JUSTA

$$P(T) = P(H) = 1/2$$

$$P(TTT \mid H_0) = \left(\frac{1}{2}\right)^3 = \frac{1}{8}$$

$P(TTT \mid HA)$ ⇒ SOB HA A MOEDA É VICIADA E NÃO SABEMOS O VALOR EXATO DE $p(T)$

ASSUMIMOS $p \sim$ UNIFORME(0,1), OU SEJA, QUALQUER VIES É IGUALMENTE PROVAVEL.

⇒ $P(TTT) = p^3$

VALOR ESPERADO

$$P(TTT \mid HA) = E[p^3] = \int_0^1 p^3 \, dp = \left.\frac{p^4}{4}\right|_0^1 = \boxed{\frac{1}{4}}$$

$$P(TTT) = \left(\frac{1}{2}\right)^3 \cdot \frac{1}{2} + \left(\frac{1}{4}\right) \cdot \frac{1}{2} = \frac{1}{16} + \frac{1}{8} = \boxed{\frac{3}{8}}$$

ILLINOIS | WEEK 1 | ASSINGMENT 1

(3) H_0 (FAIR) $= 0,66$

H_1 (NOT FAIR) $= 0,35$

TTTTT ← DATA

PRIOR		POSTERIOR
0,66	→	0,274

$P(H_0 | DATA) = ?$

POSTERIOR

$$P(H_0|DATA) = \frac{P(DATA|H_0) \cdot P(H_0)}{P(DATA)} = \frac{(1/32) \cdot (2/3)}{0,076} = 0,274$$

MARGINAL

$$P(DATA) = \underline{P(DATA|H_0)} \cdot P(H_0) + \boxed{P(DATA|H_1)} \cdot P(H_1)$$

→ $P(DATA|H_0)$ → SOB H_0 A MOEDA É JUSTA

$P(T) = 1/2$

$P(DATA|H_0) = (1/2)^5 = \boxed{1/32}$

→ $P(DATA|H_1)$ ⇒ SOB H_1, A MOEDA É VICIADA
É NÃO SABEMOS O VALOR EXATO
DE $P(T)$, (TAIL)

ASSUMIMOS $P(T) = \rho$, $\rho \sim$ UNIFORME$(0,1)$, OU SEJA
QUALQUER VIÉS É IGUALENTE PROVÁVEL

$$P(DATA) = \rho^5 \Rightarrow P(DATA|H_1) = E[\rho^5] = \int_0^1 \rho^5 dt =$$

$$E[\rho^5] = \frac{1}{6}\rho^6 \Big|_0^1 = \boxed{\frac{1}{6}}$$

→ $P(DATA) = (1/32) \cdot (2/3) + (1/6) \cdot (1/3)$
$= 0,076$

BDA → 1.4 INFERENCE ABOUT GENETIC STATUS

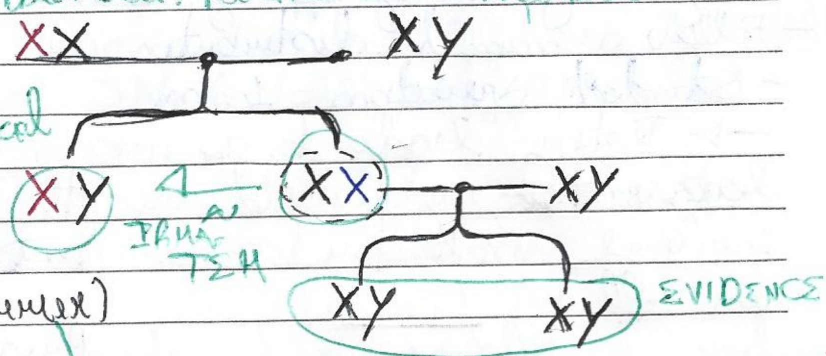
/ /

HEMOPHILIA

θ → variável não observada, parametros ou hipotese

y_i's → dados, evidências

Prior → distribuição inicial sobre θ antes de ver os dados



$\theta = 0$ (not a carrier)

$\theta = 1$ (carrier)

Data: $\vec{y} = (0,0)$

Prior: 50/50 (0,5)

PORTADORA

$Pr(\theta=1) = 0,5 \rightarrow 0,2$

$Pr(\theta=0) = 0,5 \rightarrow 0,8$

não portadora

VALORES ATUALIZADOS COM 2 FILHOS

LIKELIHOOD: $p(\vec{y}|\theta=1) = (1/2)\cdot(1/2) = (1/4)$

(VEROSIMILHANÇA)

PROBABILIDADE de gerar os dados com cada valor de Theta

$p(\vec{y}|\theta=0) = 1 \cdot 1 = 1$ (not a carrier)

MARGINAL:

$$p(\vec{y}) = p(\vec{y}|\theta=1)\cdot p(\theta=1) + p(\vec{y}|\theta=0)\cdot p(\theta=0)$$

$$= 1/4 \cdot 1/2 + 1 \cdot 1/2 = 5/8$$

POSTERIOR:

É a dist. de θ após observar os dados

$$p(\theta=1|\vec{y}) = \frac{p(\vec{y}|\theta=1)\cdot p(\theta=1)}{p(y)} = \frac{1/8}{5/8} = 1/5$$

marginal

$= 20\%$

What if there is a third child who is also XY and not affected?

TER 2 FILHOS NÃO AFETADOS REDUZ A CRENÇA DE 50% → 20%.

$y = 0$ → 3º FILHO.

→ y_1, y_2, y_3

$$p(\theta=1|y) = \frac{P(y|\theta=1)\cdot P(\theta=1)}{P(y)} = \frac{(1/2)\cdot(1/5)}{\frac{1}{2}\frac{1}{5} + 1\frac{4}{5}} = \frac{1}{9} \approx 11\%$$

ILLINOIS | WEEK 1 | ASSINGMENT 1

3 H_0 (FAIR) = 0,66
H_1 (NOT FAIR) = 0,35
TTTTT ← DATA

PRIOR → POSTERIOR
0,66 → 0,274

$P(H_0 | DATA) = ?$

POSTERIOR

$$P(H_0|DATA) = \frac{P(DATA|H_0) \cdot P(H_0)}{P(DATA)} = \frac{(1/32) \cdot (2/3)}{0,076} = 0,274$$

MARGINAL

$$P(DATA) = \underline{P(DATA|H_0)} \cdot P(H_0) + \boxed{P(DATA|H_1)} \cdot P(H_1)$$

→ $P(DATA|H_0)$ → SOB H_0 A MOEDA É JUSTA

$P(T) = 1/2$

$$P(DATA|H_0) = (1/2)^5 = \boxed{1/32}$$

→ $P(DATA|H_1)$ ⇒ SOB H_1, A MOEDA É VICIADA E NÃO SABEMOS O VALOR EXATO DE $P(T)$, (TAIL)

ASSUMIMOS $P(T) = \rho$, $\rho \sim$ UNIFORME(0,1), OU SEJA QUALQUER VIES É IGUALENTE PROVAVEL

$$P(DATA) = \rho^5 \Rightarrow P(DATA|H_1) = E[\rho^5] = \int_0^1 \rho^5 dt =$$

$$E[\rho^5] = \frac{1}{6}\rho^6 \Big|_0^1 = \boxed{\frac{1}{6}}$$

→ $P(DATA) = (1/32) \cdot (2/3) + (1/6) \cdot (1/3)$
$= 0,076$

PAIR OF DIGITAL CALIPERS

- They are generally off by -1 mm or 1 mm
- Follow a normal distribution
- Standard deviation $= 1$ mm
  - → Testing leads to a measurement of 0,5 mm larger than you would expect. In fact of this evidence, what the prob. that the calipers are off by 1 mm?

Prior

$\theta = 1: \quad y \sim N(1,1) \qquad 0,5$

$\theta = -1: \quad y \sim N(-1,1) \qquad 0,5$

LIKELIHOOD: $p(y=0,5 \mid \theta=1) = \frac{1}{\sqrt{2\pi}} \cdot e^{-\left(\frac{(y-1)^2}{2}\right)} \simeq 0,352$

MARGINAL:

$P(y=0,5) = \cancel{p(y=0,5 \mid \theta=1) \; p(y=0,5 \mid \theta=-1)}$

$= p(y=0,5 \mid \theta=1) \cdot p(\theta=1) + P(y=0,5 \mid \theta=-1) \cdot p(\theta=-1)$

$= 0,09605$

POSTERIORS

$P(\theta=1 \mid y=0,5) = \frac{0,07022}{0,09605} \approx 0,73$

0,5 → 0,73 crença atualizada

$Pr(\theta=1 \mid y=0,5) = \frac{P(\overset{0,5}{y} \mid \theta=1) \cdot P(\theta=1)}{P(y \mid \theta=1) \underset{0,5}{P(\theta=1)} + P(y \mid \theta=-1) \cdot \underset{0,5}{P(\theta=-1)}}$

~~$Pr(\theta=\pm 1)$~~

$Pr(\theta=\pm 1) = 0,5 \qquad y=0,5 \qquad y \mid \theta \sim N(\theta,1)$

BAYESIAN COMPUTATIONAL STATISTICS
WEEK 1 - GRADED ASSIGMENT 1

1) Hypotheses:

$\theta = A$: ~~..~~ you chose JAR A (70% red, 30% blue)
$\theta = B$: you chose JAR B (30% red, 70% blue)

PRIOR: ~~B~~ $Pr(\theta = A) = 0,5$
$Pr(\theta = B) = 0,5$ } JAR chosen at random

EVIDENCE: Each candy draw is blue or red, with replacement

GOAL:
Determine the minimum number of draws "n" to ensure the posterior distribution is different from the prior

$$Pr(\theta = A \mid y_1 \dots y_n) \neq 0,5$$

Since the likelihoods under both hypothesis are different, only one ~~dr~~ draw is enough.

$$Pr(\theta = A \mid y = red) = \frac{0,7 \cdot 0,5}{0,7 \cdot 0,5 + 0,3 \times 0,5} = 0,7$$

② ~~Distribution~~ Hypothesis (models) / /

- Model A $y \sim \text{Uniform}(0,1)$ $p(y|A) = 1$ Constant density
- Model B: $y \sim \text{Triangular}(0,1)$ PDF: $p(y|B) = 2y$ $y \in [0,1]$

PRIOR: $Pr(A) = Pr(B) = 0{,}5$

EVIDENCE: a sample size n, not fully specified, in the range $[0{,}25; 0{,}75]$

What is the posterior prob. of model B, given this data?

STEP 1: LIKELIHOOD FOR EACH MODEL

$Pr(y \in [0{,}25; 0{,}75] | A) = \int_{0,25}^{0,75} 1\,dy = 0{,}5$ ⇒ If you draw "n" independent draws ⇒ $(0{,}5)^n$.

$Pr(y \in [0{,}25; 0{,}75] | B) = \int_{0,25}^{0,75} 2y\,dy = y^2 \Big|_{0,25}^{0,75} = (0{,}75)^2 - (0{,}25)^2 = 0{,}5$

for "n" independent draws ⇒ $(0{,}5)^n$

STEP 2 Bayes theorem.

$$Pr(H|\text{data}) = \frac{Pr(\text{data}|H) \cdot Pr(H)}{\sum_{H'} P(\text{data}|H') \cdot Pr(H')}$$

MODEL A:

$$Pr(A|\text{data}) = \frac{Pr(\text{data}|A) \cdot Pr(A)}{Pr(\text{data}|A) Pr(A) + Pr(\text{data}|B) \cdot Pr(B)} = 1/2$$

SIMILAR TO MODEL B: $Pr(B|\text{data}) = 1/2$

④ $H_{FAIR} \Rightarrow P(\text{TAILS}) = P(\text{HEADS}) = 0{,}5$

$H_{BIASED} \rightarrow P(T) = 1$

DATA: TTTT $\Rightarrow$ 4 TAILS

PRIOR:

~~you~~ me: $P(H_F) = 0{,}99$; $P(H_b) = 0{,}01$

Friend: $P(H_F) = 0{,}5$; $P(H_b) = 0{,}5$

LIKELIHOODS:

UNDER FAIR: $P(TTTT \mid H_F) = (0{,}5)^4 = 0{,}0625$

UNDER BIASED $P(TTTT \mid H_b) = 1$

BAYES THEOREM:

$$P(H_F \mid DATA) = \frac{P(H_F) \cdot P(DATA \mid H_F)}{P(H_F) \cdot P(DATA \mid H_F) + P(H_b) \cdot P(DATA \mid H_b)}$$

ME:

$$P(H_F \mid DATA) = \frac{0{,}99 \cdot 0{,}0625}{0{,}99 \times 0{,}0625 + 0{,}01 \times 1} \cong 0{,}86$$

FRIEND:

CHANGE FROM $0{,}99 \rightarrow 0{,}86$

$$P(H_F \mid DATA) = \frac{0{,}5 \times 0{,}0625}{0{,}5 \times 0{,}0625 + 0{,}5 \times 1} \approx 0{,}059$$

CHANGE FROM $0{,}5 \rightarrow 0{,}059$

(5) ~~Setup~~ Prior $P(D) = 0{,}01$

~~Prior~~ Test

$$P(+|D) = 0{,}95 \quad , \quad P(-|D) = 0{,}05$$
$$P(+|\neg D) = 0{,}10 \quad P(-|\neg D) = 0{,}90$$

observed K positive tests

Posterior after K^+ positive tests

$$P(D|K^+) = \frac{P(D).P(+|D)^K}{P(D).P(+|D)^K + P(\neg D).P(+|\neg D)^K}$$

$$= \frac{0{,}01 \times (0{,}95)^K}{0{,}01.(0{,}95)^K + 0{,}99.(0{,}1)^K}$$

we want smallest K such that $P(D|K+) > 0{,}95$

$K = 4 \Rightarrow P(D|K+) = 0{,}988$

(6)

$P(p) \rightarrow$ pregnant $\quad P(+|P) \rightarrow$ + Test | Pregnant

$P(+|FP)$

Test $\quad P(P|+) \rightarrow$ POSTERIOR

$P(+|P) = 0{,}99 \; ; \; P(+|\neg P) = 0{,}01$

Contraception is 99% effective; $P(P) = 0{,}01$

$P(\neg P) = 0{,}99$

$$P(P|+) = \frac{P(P).P(+|P)}{P(P).P(+|P) + P(\neg P).P(+|\neg P)} \approx 0{,}5$$

7 What's the prosecutor fallacy:

$P(4 \text{ DEATHS} \mid \text{INNOCENT}) = 0{,}05$

THEN THEY JUMP TO $\Rightarrow P(\text{INNOCENT} \mid 4 \text{ DEATHS}) \Rightarrow$ SMALL

$\Rightarrow$ GUILTY

CORRECT WAY:

BAYES THEOREM: $P(\text{INNOCENT} \mid 4 \text{ DEATHS})$

$$P(\text{INNOCENT} \mid 4 \text{ DEATHS}) = \frac{P(4 \text{ DEATHS} \mid \text{INNOCENT}) \cdot P(\text{INNOCENT})}{P(4 \text{ DEATHS})}$$

IN ODD'S FORM:
$$\boxed{\text{POSTERIOR ODD'S} = \text{PRIOR ODD'S} \times \text{LIKELIHOOD RATIO}}$$

FALLACY:

$P(\text{EVIDENCE} \mid \text{INNOCENCE})$ is LOW $\rightarrow$ SO INNOCENCE IS UNLIKELY

CORRECT:

$P(\text{INNOCENCE} \mid \text{EVIDENCE})$

THEY IGNORED:

- PRIOR PROBABILITY OF GUILT (VERY LOW FOR A RANDOM NURSE)
- PROB OF 4 DEATH'S UNDER ALTERNATIVE HYPOTHESIS
- COMPARE COMPETING HYPOTHESIS.

8

H_1: The app is effective (reduces time by 50%)
H_0: the app is not effective

DATA: 12% reduction

~~the~~ POSTERIOR: $P(H_1 | DATA) = ?$

BAYES:

$$P(H_1 | DATA) = \frac{P(DATA|H_1) \cdot P(H_1)}{P(DATA|H_1) \cdot P(H_1) + P(DATA|H_0) \cdot P(H_0)}$$

PRIORS (CHOICES):

$P(H_1) = P(H_0) = 0,5$

MODEL THE % REDUCTION AS A NORMAL DISTRIBUTION

WORKS : $\sim N(50\%, \sigma_1^2)$ $\sigma_1 = \sigma_0 = 20\%$

DON'T WORK: $\sim N(50\%, \sigma_0^2)$ ~~weak~~ WEAK PRIOR

$$P(DATA|H_1) \sim EXP\left(-\frac{(12-50)^2}{2 \cdot 20^2}\right) \sim 0,16$$

$$P(DATA|H_0) \sim EXP\left(-\frac{(12-0)^2}{2 \cdot 20^2}\right) \sim 0,83$$

BAYES:

$P(H_1|DATA) = 0,164$

/ /

1)

PRIOR: $H_F \Rightarrow$ FAIR

$P(H_F) = 0{,}99$; $P(\neg H_F) = 0{,}01$

H_{BIASED} : $P(DATA | H_b) = 1^n = 1$

LIKELIHOOD:

$P(DATA | H_F) = (0{,}5)^K$

K

DATA:

HOW MANY "HEADS IN A ROW" you need to ~~____~~ your posterior that the coin is Fair drop < 15%?

BAYES THEOREM

$$P(H_F|D) = \frac{P(D|H_F) \cdot P(H_F)}{P(D|H_F) \cdot P(H_F) + P(D|H_B) \cdot P(H_B)} = \frac{(0{,}5)^K \cdot (0{,}99)}{(0{,}5)^K \cdot 0{,}99 + 1 \cdot (0{,}01)}$$

$$\frac{(0{,}5)^K \cdot 0{,}99}{(0{,}5)^K \cdot 0{,}99 + 0{,}01} < 0{,}15 \Rightarrow 0{,}5^K \cdot 0{,}99 < 0{,}15 \cdot 0{,}5^K \cdot 0{,}99 + 0{,}002$$

$$0{,}5^K \cdot 0{,}99 < 0{,}149 \cdot 0{,}5^K + 0{,}002$$

$$0{,}842 \cdot 0{,}5^K < 0{,}002$$

$$\text{Log}(0{,}5^K) < \text{Log}(0{,}002)$$

$$K \cdot \text{Log}(0{,}5) < -2{,}749$$

$$-0{,}301K < -2{,}749$$

$$0{,}301K > 2{,}749$$

$$K > 9$$

/ /

(11) PRIOR

HF = 0,5 (FAIR)

HB = 0,5 (BIASED) →

LIKELIHOOD: $P(TTT \mid HF) = (0{,}5)^3$ $P(TTT \mid HB) = 1$

SEMPRE 1

BAYES THEOREM

$$P(HF \mid TTT) = \frac{P(TTT \mid HF) \cdot P(HF)}{P(TTT \mid HF) \cdot P(HF) + P(TTT \mid HB) \cdot P(B)}$$

$$= \frac{0{,}5^3 \cdot 0{,}5}{0{,}5^4 + 1 \cdot 0{,}5} = \frac{0{,}5^4}{0{,}5^4 + 0{,}5} = 0{,}11$$

(12)

$P(D) = 0{,}01$ $P(\neg D) = 0{,}99$

D → DISEASE

TRUE POSITIVE RATE = $P(+ \mid D) = 0{,}95$

FALSE POSITIVE RATE = $P(+ \mid \neg D) = 0{,}10$

DATA: TWO POSITIVE RESULTS

LIKELIHOODS:

~~P(+,+|D) = 0,95 P(+,+|D) = 0,~~

P(2 POSITIVES | HAS DISEASE) = $0{,}95 \times 0{,}95 = 0{,}95^2$

P(2 POSITIVES | no DISEASE) = $0{,}10 \times 0{,}10 = 0{,}10^2$

BAEYES:

$$P(\text{HAS DISEASE} \mid \text{2 POSITIVES}) = \frac{P(2+ \mid D) \cdot P(D)}{P(2+ \mid D) \cdot P(D) + P(2+ \mid \neg D) \cdot P(\neg D)}$$

$$= \frac{0{,}95^2 \cdot 0{,}01}{0{,}95^2 \times 0{,}01 + 0{,}1^2 \cdot 0{,}99} = \frac{0{,}009}{0{,}009 + 0{,}1^2 \times 0{,}99} \cong 0{,}47$$

/ /

(15) SETUP

D → has Disease

¬D → Does not have

+ → positive result

Prior: $P(D) = 0{,}01 \quad P(\neg D) = 0{,}99$

TRUE POSITIVE RATE ⇒ $P(+|D) = 0{,}95$

FALSE POSITIVE RATE ⇒ $P(+|\neg D) = 1 - 0{,}90 = 0{,}10$

Bayes: $P(D|+) = \dfrac{P(+|D) \cdot P(D)}{P(+|D) \cdot P(D) + P(+|\neg D) \cdot P(\neg D)}$

$= \dfrac{0{,}95 \cdot 0{,}01}{0{,}95 \times 0{,}01 + 0{,}10 \times 0{,}99} \approx 0{,}08$